

Trainings ▾

Preparatory Steps

- Remove the alternator belt. Refer to Procedure 013-005 in Section 13. (</qs3/pubsys2/xml/en/procedures/89/89-013-005.html>)
- Remove the alternator drive pulley, if necessary. Refer to Procedure 009-010 in Section 9. (</qs3/pubsys2/xml/en/procedures/89/89-009-010.html>)

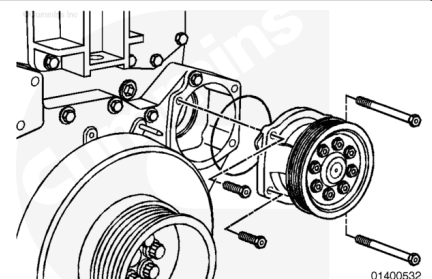


Remove

Remove the two long and the two short bolts from the alternator drive. Remove the alternator drive and o-ring.

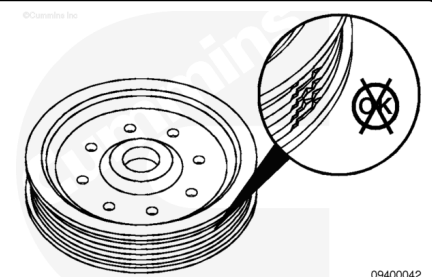
Discard the o-ring.

Note : Some components shown removed for clarity.



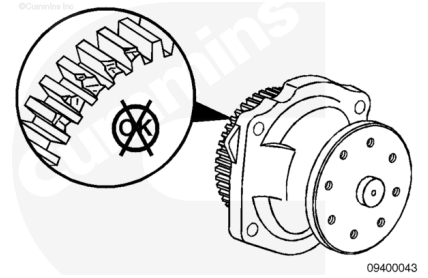
Initial Check

Inspect the pulley for nicks, cracks, or excessive wear in the belt grooves or other damage.



Inspect the alternator drive gear for cracks, chipped, or missing teeth.

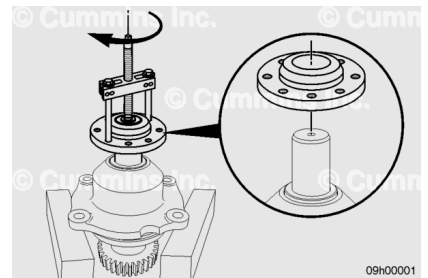
Inspect the alternator drive housing for cracks and confirm that the shaft turns smoothly.



Disassemble

Use a standard puller, Part Number ST-647, or equivalent.

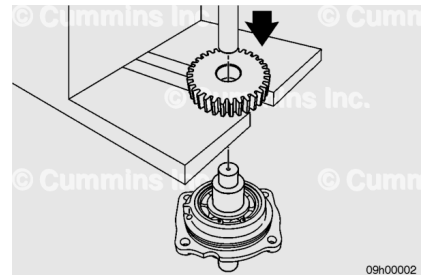
Remove the alternator drive pulley hub.



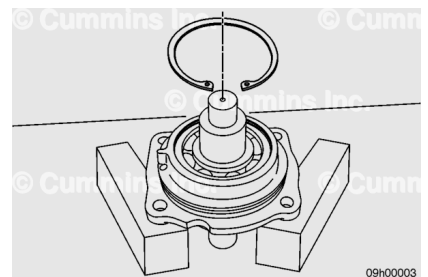
Use a water pump bearing separator, Part Number 3375326, or equivalent, to support the gear.

Use an arbor press to remove the shaft and alternator drive assembly out of the gear.

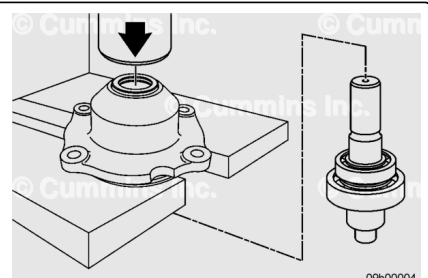
Remove the thrust plate from the shaft.



Remove the retaining ring from the back side of the housing.

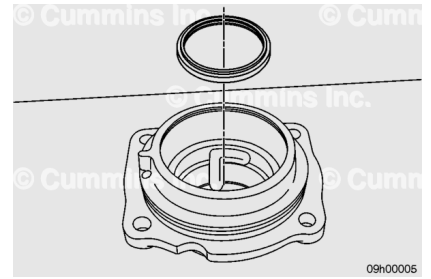


Use an arbor press to remove the shaft and bearing assembly out of the housing.



Remove the oil seal from the housing.

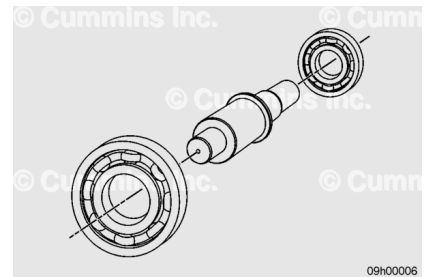
Discard the oil seal.



Use an arbor press to remove the shaft from the bearings.

Note : The smaller bearing may remain in the housing when the shaft is removed. If necessary, press the bearing from the housing.

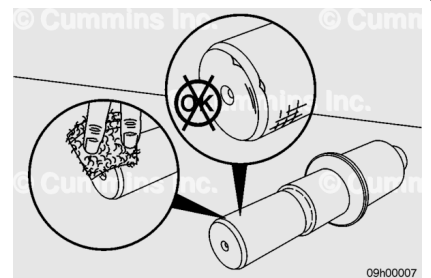
Discard the bearings.



Inspect for Reuse

Clean and inspect the end of the shaft.

Use emery cloth to remove any burrs or debris caused by removal of the impeller, drive coupling, or pulley hub.



Measure the press fit for drive gear, alternator pulley hub, and fuel pump drive coupling.

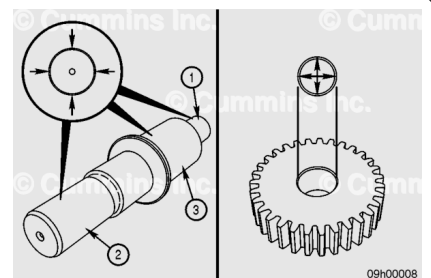
Measure the outside diameter of the shaft in each location:

1. Drive gear
2. Alternator hub location
3. Fuel pump drive coupling.

Measure the inside diameter of the mating part.

Use the following to determine the press fit.

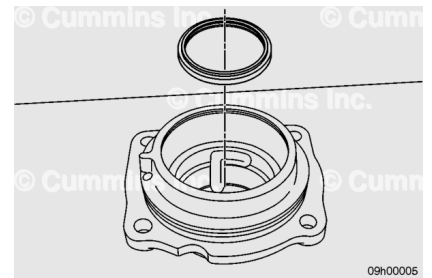
Press fit = shaft outside diameter (1) minus the gear inside diameter.



mm		in
0.02	MIN	0.0008
0.06	MAX	0.0024

Assemble

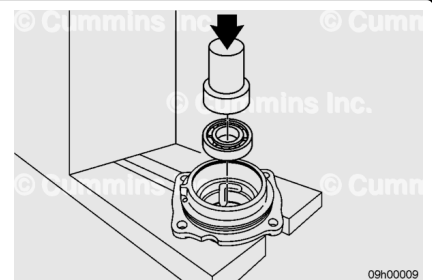
Install the oil seal into the housing.



⚠ CAUTION ⚠

Be sure that the mandrel contacts only on outer race of bearing. Contact with inner race can cause bearing damage.

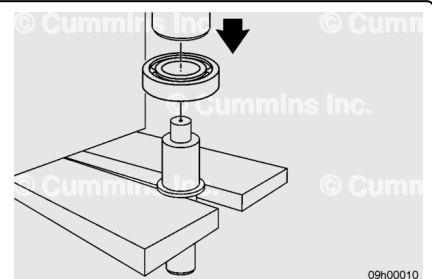
Use an arbor press and a mandrel to install a new small bearing into the housing.



⚠ CAUTION ⚠

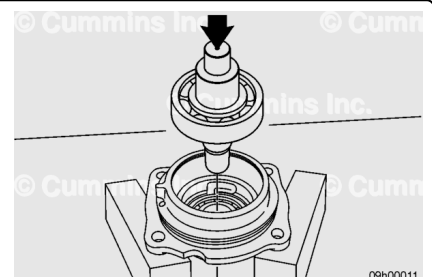
Be sure that the mandrel contacts only on inner race of bearing. Contact with outer race can cause bearing damage.

Use an arbor press and a mandrel to install a new large bearing onto the shaft. Be sure that the bearing is pressed flush against the shoulder of the shaft.



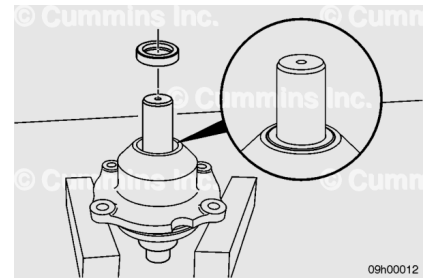
Install the shaft assembly into the housing.

If necessary, lightly press the shaft assembly into the housing. Be sure the shaft is in line with the housing as **not** to damage the bearings.



Install the oil seal into the housing.

Install the seal so that it is flush with the outside of the housing.

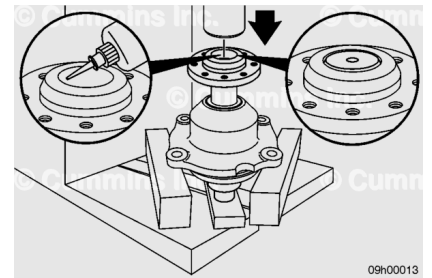


Support the shaft.

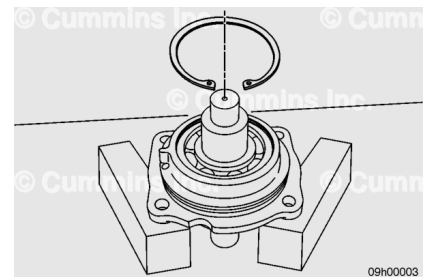
Apply Loctite™ 609, Part Number 3823718, or equivalent, to the inside diameter of the hub.

Use an arbor press to install the pulley hub onto the housing.

Install the hub so that it is flush with the end of the shaft.



Install the retaining ring for the large bearing.

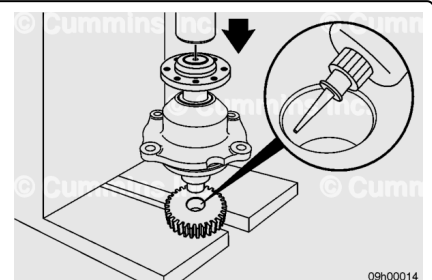


Install the thrust plate onto the shaft.

Apply Loctite™ 609, Part Number 3823718, or equivalent, to the gear inside diameter.

Support the gear in a press.

Press the shaft onto the gear.



Install

Confirm that the alternator to fuel pump drive coupling is in place.

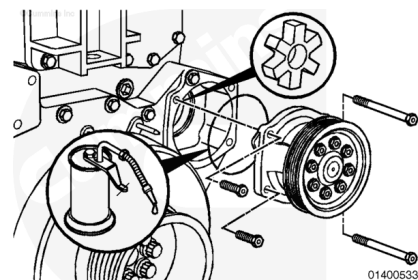
Lubricate the alternator drive o-ring with clean engine oil and install onto the alternator drive assembly.

Install the alternator drive assembly.

Install the two long alternator drive capscrews on the right side of the assembly.

Install the two short capscrews on the left side of the assembly.

Torque Value: 113 n•m [83 ft-lb]



Finishing Steps

- If removed, install the alternator drive pulley. Refer to Procedure 009-010 in Section 9. (</qs3/pubsys2/xml/en/procedures/89/89-009-010.html>)
- Install the alternator belt. Refer to Procedure 013-005 in Section 13. (</qs3/pubsys2/xml/en/procedures/89/89-013-005.html>)
- Start the engine and check for proper operation.

